

ADVANCED OPTICAL MATERIALS

Supporting Information

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Anti Stokes Thermometry of Plasmonic Nanoparticle Arrays

Simone Ezendam, Lin Nan, Ianina L. Violi, Stefan A. Maier, Emiliano Cortés, Guillaume Baffou* and Julian Gargiulo**

Supplementary Information to "Anti-Stokes Thermometry of Plasmonic Nanoparticle Arrays"

Simone Ezendam,[†] Lin Nan,[†] Ianina L. Violi,[‡] Stefan A. Maier,^{¶,§,†} Emiliano Cortés,^{*,†} Guillaume Baffou,^{*,||} and Julian Gargiulo^{*,†}

[†]*Nanoinstitute Munich, Faculty of Physics, Ludwig-Maximilians-Universität, München, Germany*

[‡]*Instituto de Nanosistemas, Universidad de San Martín, CONICET, Buenos Aires, Argentina*

[¶]*School of Physics and Astronomy, Monash University, Clayton 3800, Australia*

[§]*Department of Physics, Imperial College London, London SW7 2AZ, UK*

^{||}*Institut Fresnel, CNRS, Aix Marseille Univ, Centrale Marseille, Marseille, France*

E-mail: emiliano.cortes@lmu.de; guillaume.baffou@fresnel.fr; jgargiulo@unsam.edu.ar

S0 Materials and methods

S0.1 Fabrication

The samples were fabricated with a standard topdown fabrication method. The masks were written with electron beam lithography on 120 nm 950K PMMA with 20 kV acceleration voltage. Subsequently, the mask was developed in isopropanol-MIBK developer and deposited with 40 nm gold with electron beam evaporation.

S0.2 Optical characterization

Dark-field images, scattering spectra, and transmission spectra were taken using a WITec microscope (Germany). Transmission spectra were acquired with a 20x water immersion objective (Zeiss) by illuminating the sample from the bottom with a collimated beam. The transmission spectra were normalized by the lamp spectrum. Single-particle dark-field scattering spectra were acquired using a transmission configuration employing a condenser and 63x water immersion objective (Zeiss). The dark-field scattering spectra were normalized by subtracting the background and then dividing by the background.

S0.3 Scanning electron microscopy

Scanning electron microscopy (SEM) images were acquired with an FE-SEM (Zeiss) operated at 20 kV using an SE2 detector. In preparation, the samples are spin coated with a conductive polymer layer (ESpacer 300Z).

S0.4 Anti-Stokes photothermal measurements

Data was acquired with a 60x water immersion objective (Olympus) on a Nikon Ti-Eclipse inverted microscope, equivalent to a 67x magnification. A 647 nm CW laser (OBIS) was focused on the backfocal plane of the objective to achieve wide-field illumination. With fluorescent beads we determined the beam size to be around 18 μm (FWHM). Power calibration was performed using a powermeter slide (Thorlabs) and the maximum irradiance was calculated for the center of the beam. The collection fiber has a core size of 400 μm (TP02815630). The signal was filtered transmitted to a spectrometer (Princeton instruments, SP-2-500i) with a 50 g/mm grating and Pixis 100B camera. The integration time was 30s. The background was acquired in the same way, in a position without array. We tried to account for a bleaching of the background by keeping the ratio between data and background spectrum in the region from 770 nm to 800 nm constant, using the lowest power as a refer-

ence. This decision was guided by the raw data for d065 and d100, because the raw intensity in this region is identical to the background spectrum. Here, scaling the background to a constant ratio is the same as scaling the background in this region to be equal to the raw spectrum. Due to the redshifted photoluminescence spectrum of d157 the signal from 770 nm to 800 nm is dominated by the antenna itself and the scaling does hardly effect the background correction.

S0.5 QLSI photothermal measurements

The setup described above was modified slightly to allow a quasi-simultaneous measurement using QLSI thermometry. A Köhler illumination was added in transmission and the signal was instead sent to a CCD with a modified hartmann grating (Phasics), again first filtered by a rejection filter 647 nm.

S1 Influence of choice of distribution

In Figure S1, three temperature-dependent distribution of the states that provide the extra energy for anti-Stokes emission $n(\lambda, \lambda_{exc}, T)$ are evaluated. Figure S1a, shows the ratio Q_{i,j_N}^{AS} as defined in equation (4) of the main article, corresponding to a value of $\beta = 900 \mu\text{m}^2 \text{K mW}^{-1}$. Three distribution ratios are shown, Fermi-Dirac, Bose-Einstein, and Boltzmann. Figure S1b shows the histograms for the measured β of a d100 array, analyzed using the three different distributions functions. While the distributions change, the median value remains almost unaffected. Figure S1c shows the temperature increase versus excitation irradiance, using the median values of the three distributions of Figure S1b. From the three figures, it can be concluded that the choice of distribution has a very minor effect on the results.

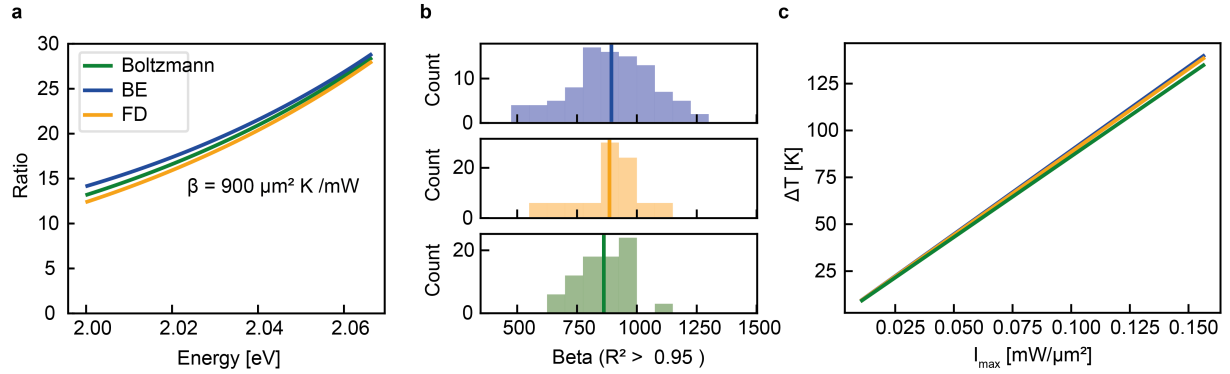


Figure S1: **Influence of the distribution.** (a) Simulated anti-Stokes ratios under the same conditions with only a change in the choice of distribution (Boltzmann, Bose-Einstein and Fermi-Dirac) (b) Fitted photothermal coefficient to the same three d100 arrays using the different distributions. (c) Temperature increase against irradiances from the median photothermal coefficient from (b), showing that the choice of distribution has a very minor effect.

S2 Thermal modelling and AS thermometry Figures of Merit

S2.0.1 Single particle

For a single NP, the total emitted PL emission P_{PL}^s is given by

$$P_{PL}^s = I_{exc} \sigma_{abs} \phi \quad (1)$$

The temperature increase of a single particle T_0^S situated at the center of a Gaussian beam illumination can be estimated as

$$\Delta T_0^S = \frac{\sigma_{abs} I(0)}{4\pi \bar{\kappa} R_L} = \frac{\ln(2) \sigma_{abs} P}{\pi^2 H^2 \bar{\kappa} R_L}. \quad (2)$$

Here, σ_{abs} is the absorption cross section, $I(0)$ is the irradiance at the center of the beam, P is the power measured at the sample position, H is the full-width-half-maximum (FWHM) of the Gaussian beam, $\bar{\kappa}$ is the averaged thermal conductivity of the surrounding medii and R_L is the Laplace radius¹ that takes into account the geometry of the individual nanoparticle. Combining equations 1 and 2 yields

$$F^s = \phi 4\pi \bar{\kappa} R_L \quad (3)$$

S2.0.2 Arrays of particles

For an array of identical NPs,

$$P_{PL}^a = N P_{PL}^s \quad (4)$$

The general dependence of ΔT_0^{ext} with the number of particles N for a 2 dimensional array is given by,²

$$\Delta T_0^{ext} \propto \Delta T_0^S \frac{R_{PL}}{p} \sqrt{N} \quad (5)$$

Dividing equations 4 and 5 yields

$$F^{2D} \propto \frac{\sqrt{NA}}{R_L} F^s \quad (6)$$

where A is the area of the unit cell (for a square lattice, $A = p^2$, with p the particle distance).

For a Gaussian illumination of an infinite square array, the collective temperature increase at the center of the illumination can be estimated as³

$$\Delta T_0^{ext} \approx \frac{\sigma_{abs} P}{\bar{\kappa}} \sqrt{\frac{\ln(2)}{4\pi}} \frac{1}{HA} \left(1 - \frac{4\sqrt{\ln(2)A}}{\pi H} \right). \quad (7)$$

If only the emission from an area with a diameter D as illustrated in Figure 1b of the main article is detected, the total PL emission from the collection area is

$$P_{PL}^a = I_{exc} \sigma_{abs} \phi N = \frac{I_{exc} \sigma_{abs} \phi \pi D^2}{4A} \quad (8)$$

Combining equations 8 and 7 yields

$$F^a = \frac{2\sqrt{\ln 2} \pi \bar{\kappa} D^2 \phi}{H} = \frac{D^2}{H} \frac{1}{2\sqrt{\pi} R_L} F^s \quad (9)$$

where we have assumed that $H \gg p$ and therefore neglected the second term in the parenthesis of equation 7

S3 Sample size characterization using Scanning electron microscopy (SEM)

Figure S2 shows a full exemplary array (d100). The total size of all arrays is 40 μm . For d065 and d100, the pitch is 250 nm, while for d157 it is 350 nm, as shown in Figure S3.

The diameter of the three different disk sizes was characterized by measuring the area of

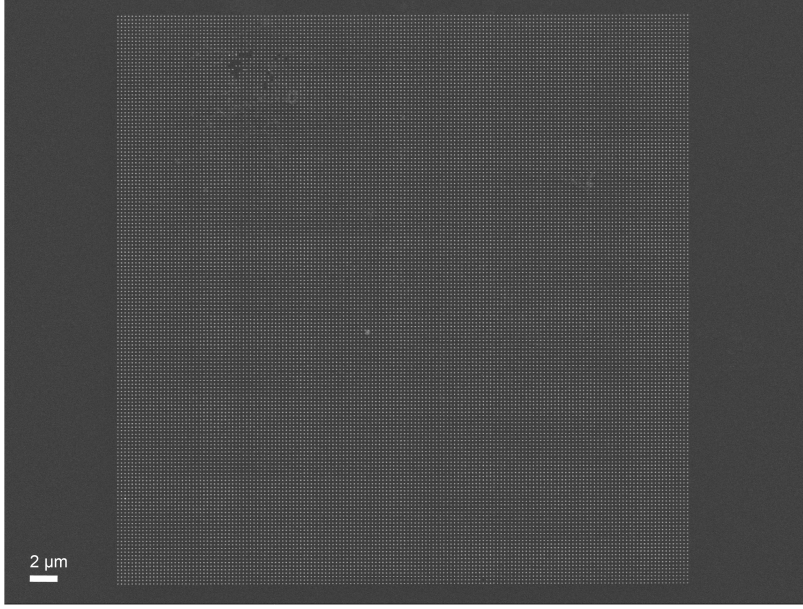


Figure S2: **Overview of an entire array.** SEM image of an exemplary d100 array.

many equivalent disks. Three representative images for the arrays are shown in Figure S3. The median diameters are $65.3 \text{ nm} \pm 0.7 \text{ nm}$ ($\sigma = 3.3 \text{ nm}$), $100.1 \text{ nm} \pm 0.5 \text{ nm}$ ($\sigma = 3.0 \text{ nm}$) and $157.0 \text{ nm} \pm 1.0 \text{ nm}$ ($\sigma = 4.0 \text{ nm}$)

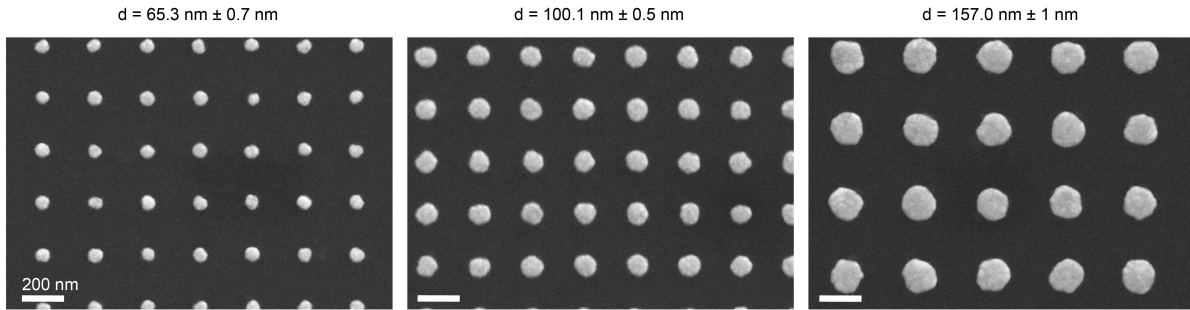


Figure S3: **Sizes of the disks.** SEM images of all three types of arrays showing the different sizes and geometries.

S4 Annealing

To investigate the photothermal properties of the arrays, we annealed them first at maximum power for 2 min to stabilize the signal. We observed a change in in transmission on the QLSI

images. To investigate this further, we acquired transmission spectra on illuminated and non-illuminated arrays. As can be seen in Figure S4, there is significant blue-shift, suggesting that there is a change in the disks. In addition to the arrays, the samples also have isolated

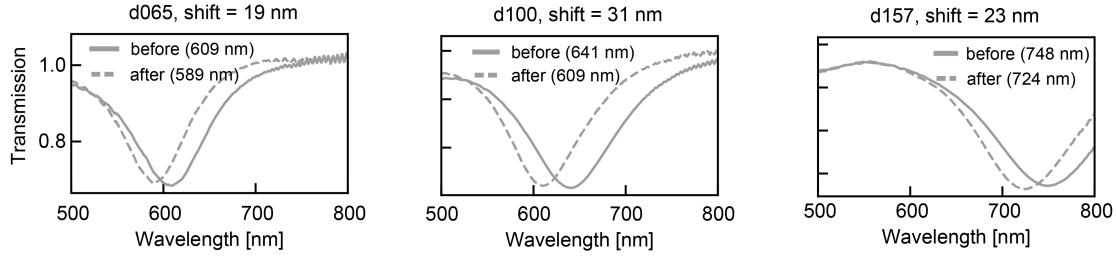


Figure S4: **Changing transmission spectra.** The transmission spectra for all three sizes show a blue-shift in the transmission after annealing.

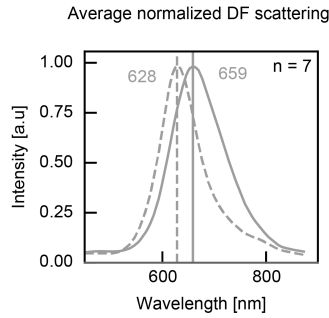


Figure S5: **Changing dark field scattering spectra.** Average normalized dark-field spectrum of 7 single disks before (solid line) and after (dashed line) annealing.

single disks. Figure S5 shows the spectrum of the single disks before and after annealing. Upon illumination of the single disks with a focused 633 nm laser with 10 to 20 mW, we witnessed a quick blue shift that did not change upon further illumination. This blue shift was again on the order of 30 nm, similar to the shift in the transmission of the arrays. Even though the resonances of the antennas appear to be shifting 20 - 30 nm upon annealing, no change of shape is apparent in the SEM images. For example, we determined the average diameter of two areas of d100 arrays with and without annealing, which were identical with $99.7 \text{ nm} \pm 0.5 \text{ nm}$ and $100.1 \text{ nm} \pm 0.6 \text{ nm}$. An explanation for a change of the optical properties without a change in geometry is a change in crystallinity, as reported recently.⁴

S5 Occurrence of a delocalization regime

The temperature increase of a given nanoparticle now has two contributions: First ΔT^S , the increase in temperature of a single particle, and second ΔT^{ext} delivered by the surrounding nanoparticles. A delocalization regime occurs when the collective heating is higher than the individual heating. The occurrence of a localization vs. delocalization regime can be predicted by considering a dimensionless parameter ζ_2 that estimates the ratio $\Delta T_{0,s}/\Delta T_{0,ext}$.

$$\zeta = \frac{3HR_L}{p^2} \quad (10)$$

S6 Temperature variation inside the collection volume

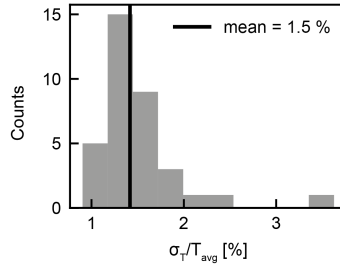


Figure S6: **Temperature variation inside the collection volume.** Histogram of the temperature variation inside the collection volume calculated as the standard deviation divided by the average Temperature.

S7 Optical simulations of disks

The simulated dimensions are as measured with the SEM (see section S3) in $D = 65$ nm (d065), $D = 100$ nm (d100), $D = 157$ nm (d157), with heights of 40 nm.

The optical properties of the nanodisks were investigated using FDTD simulations (ANSYS Lumerical Software ULC, Release 2022 R1, Version 8.27.2898). The gold disks (Johnson & Christy) were 70 nm, 100 nm and 150 nm in diameter, and the height was 40 nm for all. To simulate an aqueous medium the surrounding medium was set to a refractive index of

1.333. The substrate was simulated as a rectangular prism consisting of Glass (Palik). To calculate absorption and scattering cross-sections a TFSF source was used. The injection was simulated along the z-axis. The absorption cross-section is calculated using an analysis group inside the TFSF source. The analysis group consists of six 2D monitors that form a closed box and measure the net power flowing in/out of the box. Similarly, the scattering cross-section is calculated by an analysis group outside the TFSF source. Since this simulation has a plane of symmetry in x and in y dimension, the simulation time was reduced by setting the x min boundary condition to anti-symmetric and the y min boundary condition to symmetric. A mesh override region with $d = 1$ nm in a region of 250 nm from the center was used in all dimensions. The simulated optical properties of the single disks are shown in Figure S7.

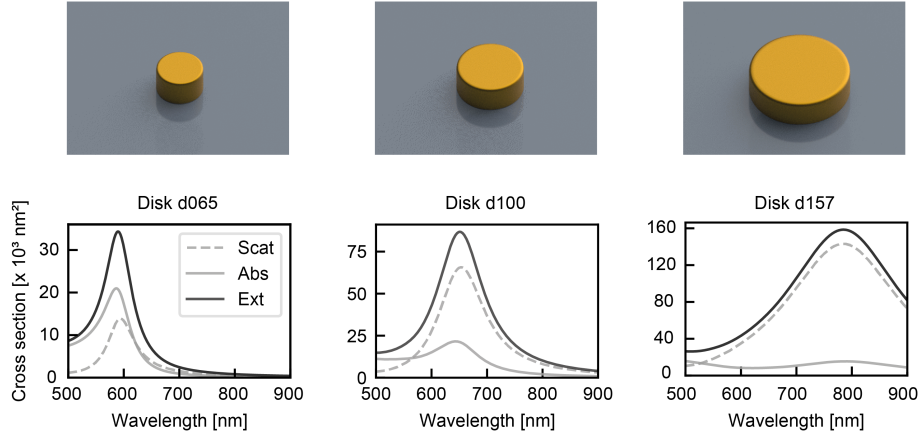


Figure S7: **Optical simulations of single disks.** Simulated scattering, absorption and extinction cross sections of disks with 65, 100, and 157 nm diameter and 40 nm height.

S8 Expected temperatures of the measured arrays

The single and collective heating can be calculated using Equations 2 and 7. The values of the absorption cross sections at 647 nm were taken from simulated optical properties (see section S7) and the Laplace radii R_L calculated according to S9. The values used are shown in Table S1. The average thermal conductivity of the media is $\bar{\kappa} = 0.8 \text{ W m}^{-1} \text{ K}^{-1}$.

Using these values, the estimated β^s for the isolated nanodisks are $\beta_{d065}^s = 10 \text{ } \mu\text{m}^2 \text{ K mW}^{-1}$, $\beta_{d100}^s = 47 \text{ } \mu\text{m}^2 \text{ K mW}^{-1}$ and $\beta_{d157}^s = 13 \text{ } \mu\text{m}^2 \text{ K mW}^{-1}$ at the excitation wavelength of 647 nm.

Using an experimental determination of H and P , we retrieve the expected temperature at the center of the illumination. The results are shown in Figure 4b in the main text (dashed lines). Although the trends are predicted reasonably well, the temperatures for d100 are overestimated by more than a factor of two, while measurements with QLSI and anti-Stokes (AS) thermometry match well. This might be caused by an overestimation of the absorption cross-section. The simulations shown in Figure S7 were performed under the assumption of a perfect disk with the desired size. The estimations could most likely be improved if the true absorption cross section was known.⁵

Table S1: Values used for the theoretical estimation

	d065	d100	d157
$R_L[\text{nm}]$	33.24	45.77	65.39
$\sigma_{abs} [\text{nm}^2]$	3252.75	21514.50	8473.73
$p [\text{nm}]$	250	250	350

S9 Calculation of Laplace Radius

The Laplace Radius of a NP is given by,¹

$$R_L = R_{eq}\alpha \quad (11)$$

Where R_{eq} is the radius of a sphere with the same volume as the NP. For a disk of diameter d and height h , it can be calculated as

$$R_{eq} = \left(\frac{3}{16} d^2 h \right)^{\frac{1}{3}} \quad (12)$$

α is a factor that for disks is given by

$$\alpha = \exp \left(0.04 - 0.0124 \log\left(\frac{d}{h}\right) + 0.0677 \left(\log\left(\frac{d}{h}\right)\right)^2 - 0.00457 \left(\log\left(\frac{d}{h}\right)\right)^3 \right) \quad (13)$$

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